

Autonomic Software and Systems assignment 2

Task 1

1. The main steps presented are: Detecting the state of the system, selecting a control action and receiving the reward signal.
Detecting the state of the system is done by observing the state of the program at discrete time steps. It measures the execution time until the current time point and decides a state for it.
Selecting the control action is done by selecting a tuning operation for the time delay for the next running function block.
Receiving the reward signal is after the action is applied, when the system moves to a new state. The controller computes a reward signal based on how good the response time is. It then updates the Q-table with this value.
2. Other methods are model-based approaches, architecture-based approaches and control theory. Model-based approaches support real-time monitoring and reasoning with system, goal and environment models. Architecture-based approaches monitor the system with an external adaptation layer. Control theory is only used in real-time applications and is based on comparing the output to a desired value.
3. In the paper there were several suggestions on ways to improve it. They say using multiple learning trials is needed for it to stabilize and suggest using the approach in soft real-time systems in the incremental learning form. They also say that future work will focus on improving the training time and precision, and they plan to do this by modelling the state space as a fuzzy state space and by using cooperative agents to speed up the learning.

Task 2

1. In the reference approach the adaptation is realized with internal mechanisms, so the logic is integrated inside the components. The external MAPE approach on the other hand is realized by adding a loop outside of the managed system.
2. Yes, since it reduces the complexity of the control-flow, it becomes easier to understand. It also reduces the fault density which means that fewer faults happen relative to the functionality.
3. A real-life example would be a load-balancer, such as the ones we talked about last week. It is an example since it monitors the system, most often the CPU utilization and automatically starts new instances if the load gets too high. It also closes instances in case the load decreases. It is a loop since it monitors the system and adapts based on the data of the system.